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Paper Code : 21324

F- 424

B.C.A. (Fifth Sem.) Examination, 2025-26

(New Course)

Paper No - BCA-504

Software Engineering

Time : Three Hours] *[Maximum Marks : 70*

Note : Attempt any **five** questions. **All** questions carry equal marks. Symbols used have their usual meanings.

1. (a) What are "Software crises"? Discuss the evolution of software.

   (b) Explain the term "SDLC" specifying its role in software industry.

2. (a) What do you mean by spiral model of software development? Write down its pros and cons.

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   (b) Write down various ways of gathering requirements of customers. Give the detailed structure of SRS.

3. (a) Illustrate the features of a good software & explain each.

   (b) How cohesion is different from coupling? What are various types of cohesion.

4. (a) Describe different types of design approaches.

   (b) How modularity effects any design? Justify with one example.

5. (a) Explain "CASE Tools" in details.

   (b) What are various phases of system investigation?

6. (a) Elaborate "PERT" i.e. Program evaluation review technique in context with software development considering any relevant example.

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(b) How "Black-Box" testing is different from "White-box" Testing? Explain their types.

7. (a) Differentiate software engineering and software re-engineering.

(b) Discuss various types of maintenance. Why this phase is assumed to be the longest phase in SDLC models? Justify your answer.

8. Write in brief about any **four** of the following:

(a) Acceptance Testing

(b) Feasibility study

(c) Prototyping 4GL tools

(d) Test cases

(e) COCOMO method

(f) Project Management

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Paper Code : 21324

F-424

B.C.A. (Fifth Semester) Examination, 2023-24

(New Course)

Paper - BCA-504

SOFTWARE ENGINEERING

Time : Three Hours] *[Maximum Marks : 70*

Note : Attempt any **five** questions. **All** questions carry equal marks. Symbols used have their usual meanings.

1. What is SRS? Explain the benefits of SRS. Develop SRS for Payroll Management System. Follow IEEE SRS format.

2. (a) Briefly explain any two problems related to software development.

(b) Explain the different categories of software maintenance.

3. (a) What is function oriented design? Explain its main element and features.

(b) What is software testing? Explain the need of system testing. Also describe the process of system testing.

4. (a) What is waterfall model? Explain whether this model is suitable for developing a Railway Reservation System or not.

(b) Explain the importance of DFD. Draw various symbols of DFD with the significance of each.

5. (a) What is meant by software testing? Explain black box and white box testing.

(b) What is software engineering? What is software crisis? How can it be managed?

6. (a) Differentiate between process and product. Which one is more important and why?

   (b) Define 4GL's. Discuss how 4GL's help to solve the complex problem. Also list its limitations.

7. (a) Define CASE tools. List four broad categories of them and briefly explain any two of them.

   (b) Draw PERT chart for the development of an "Online Student Admission System".

8. Write short notes on the following.

   (a) Prototyping model

   (b) Regression testing

   (c) Data Abstraction

   (d) Maintainability and Reusability.

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Paper Code : 21324

F-424

B.C.A. (Fifth Sem.) Examination, 2022-23

(New Course)

Paper-BCA-504

SOFTWARE ENGINEERING

Time : Three Hours] ***[Maximum Marks : 70***

Note : Attempt any **five** questions. **All** questions carry equal marks. Symbols used have their usual meanings.

1. (a) Describe the problem known as "Software-Crisis".

   (b) "A software does not wear out". Justify this statement clearly.

2. (a) What is "SDLC"? Explain the spiral model of software development.

   (b) How to develop "SRS-document"?

3. (a) Illustrate qualities of any good software in detail.

   (b) What are the data structure oriented methods of requirement analysis?

4. (a) Write in brief about the following terms :

   (i) CASE Tools

   (ii) PERT

   (b) Write down the general principles of system investigations.

5. (a) What do you mean by "DFD"? Draw a DFD for "Tic-Tac-Toe" game.

   (b) Describe the concept of test-cases. How to carry out Black-Box-Testing?

6. (a) What is reliability? Why it is necessary for any software to be of high-reliability?

   (b) How end-user computing evolved? Explain.

7. (a) Draw the life-cycle of an information system and describe in detail.

(b) How verification is different from validation.

8. (a) Write down the role of software engineer. Describe various paradigms of software engineering.

(b) Write short notes on any **two** of the following :

(i) COCOMO - model

(ii) Types of testing

(iii) Software Maintenance

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Total No. of Questions : 8] [Total No. of Printed Pages : 3

Paper Code : 21324

F-424

B.C.A. (Fifth Semester) Examination, 2021

(New Course)

Paper-BCA-504

SOFTWARE ENGINEERING

Time : Three Hours] *[Maximum Marks : 70*

Note : Attempt any five questions. All questions carry equal marks. Symbols used have their usual meanings.

1. (a) What is the principal aim of the software engineering discipline? What does the discipline of software engineering discuss?

(b) Explain software development life cycle model with example.

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2. (a) Explain how do the use of software engineering principles help to develop software product cost-effectively and timely.

(b) What are the other component process in a software process? Describe the purpose of each.

3. (a) Differentiate between functional requirement and non functional requirement.

(b) Take some quality attributes for the SRS and try to define some metrics for Quantifying.

4. (a) What is maintainability? Explain with reusability.

(b) What is modularity? Describe principle of software engineering.

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5. (a) What is the life cycle of an information system? Explain phase of system.

   (b) Explain fourth generation language with example.

6. (a) What is the limitation of GLS? Explain impact of 4GLS.

   (b) Describe major influencing factors.

7. (a) What is case tools? Explain with deft case tools.

   (b) Describe system testing in details.

8. Write short notes on any four:

   (a) Database

   (b) Robustness

   (c) Relational database

   (d) PERT

   (e) System review

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Total No. of Questions : 8] [Total No. of Printed Pages : 3

Paper Code : 21324

F–424

B.C.A. (Vth Semester) Examination, 2019–20

(New Course)

SOFTWARE ENGINEERING

Paper-BCA-504-N

Time : 3 Hours] [Maximum Marks : 70

Note :– Attempt any *five* questions. All questions carry equal marks. Symbols used have their usual meaning.

1. (a) Define software engineering. Explain various software myths and applications.

   (b) Explain the various stages of software development in detail.

2. (a) What is E-R diagram ? Write down the various notations used in ER diagram. What are the various purposes of ER diagram in software development ?

   (b) Describe the major components (with their corresponding symbol) of DFD and also explain the various types of DFD with example.

3. (a) Explain the function oriented and object oriented approaches for software design.

   (b) What is requirement analysis ? Briefly explain different types of requirement problems that should be identified and resolved during requirement analysis activity.

4. (a) What is cohesion and coupling ? Write the differences between them.

   (b) Explain the COCOMO model in detail with the help of various equations. Compare the basic COCOMO model with the detailed COCOMO model.

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5. (a) What is software testing ? Explain its various objectives and principles.

   (b) Explain the functional testing and differentiate it with structural testing.

6. (a) Describe the role of quality control and quality assurance in software quality.

   (b) Describe the SEI CMM model with its different levels in detail.

7. (a) Explain reverse engineering in detail. Write the difference between reverse engineering and re-engineering.

   (b) Explain 4GL tools in detail.

8. Explain any *four* of the following :

   (i) SRS with characteristics

   (ii) Alpha & Beta Testing

   (iii) Verification Vs. Validation

   (iv) ISO 9001

   (v) CASE Tools

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Total No. of Questions : 8] [Total No. of Printed Pages : 2

Paper Code : 21324

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B.C.A. (Fifth Semester) Examination, 2018

(New Course)

Paper-BCA-504

SOFTWARE ENGINEERING

Time : 3 Hours] [Maximum Marks : 70

Note :– Attempt any *five* questions. All questions carry equal marks. Symbols used have their usual meanings.

1. (a) What are the symptoms of the present software crisis ?
   (b) Explain water fall model with suitable example.
2. (a) Describe the relationship of software engineering to other areas of computer science.
   (b) Which phase consumes the maximum effort for Developing a typical software product ?
3. (a) List the five desirable characteristics of good software requirement specification (SRS) document.
   (b) Explain formal specification technique.

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4. (a) What is case study ? Explain case tools.
   (b) Describe fourth and fifth generation of computer science.
5. (a) What is the life cycle of an information system ? Explain Phase of system.
   (b) What are the general principles of system investigation ?
6. (a) Explain program evaluation review technique (PERT).
   (b) What is system testing ? Explain system review.
7. (a) What is Quality ? Explain the Software assurance.
   (b) Explain all factors affecting software.
8. Write short notes on any *four* :
   (a) Operating system
   (b) Maintainability
   (c) System investigations
   (d) Perception of a problem
   (e) Reusability

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